

"Pick the red tomato" — a robot that obeys a sentence

RAISE 2026 · Day 2 Demo

Day 2 demo, captured LIVE from the simulator · RAISE 2026 Summer School, ENET'Com Sfax



Everything on the next slides is a real capture: the Gazebo window, the terminals, the model outputs.

Step 1-2 · Start the world

Two terminals, one command each — then leave them alone

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```
terminals 1 + 2
$ sim_d2      # terminal 1 - the greenhouse world (plant-row parking)
$ grasp_d3    # terminal 2 - the grasp service (it IS the grasp)
```

The robot parks facing a tomato plant — the exact spot the model was trained at (deterministic parking).

Step 3 · Install the brain

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A fine-tuned SmolVLA checkpoint — one command, resumable, public

```
terminal 3

$ get_brain_d4
Downloading the reference brain (687 MB) – interruptions RESUME...
✓ reference brain installed – run:  vla_d4 --task C --spawn

# installs to ~/raise_checkpoints/smolvla_C_ref/checkpoints/006000/
# pretrained_model/ = model.safetensors (450M params) + processors
# skip it? first run auto-downloads from Hugging Face:
#   scalexi/smolvla-raise2026-ripeness-ref  (public, no key)
```

Trained in ~1.5 h on 49 auto-recorded demonstrations. All Day-2 tools auto-detect it.

Step 4 · Look inside ONE brain call

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Three inputs, seven numbers out — that is the whole secret

terminal 3

```
$ vla_one_d4 --spawn
```

A VLA is a function: `action = model(image, instruction, state)`

INPUT 1 — image : 224×224 RGB from /wrist_camera/image_raw (what it SEES)

INPUT 2 — words : "pick the red tomato" (what we WANT)

INPUT 3 — state : 7 floats from /joint_states (where it IS)

OUTPUT — action : 6 joint targets + gripper (what to DO next) [163 ms]

joint	state (in)	action (out)	move?
shoulder_pan	+0.002	+0.058	↑ up
shoulder_lift	-1.560	-1.524	↑ up
elbow	+1.590	+1.537	↓ down
wrist_1	-1.570	-1.638	↓ down
gripper	+0.054	+0.003	↓ down

The full executor is exactly this call, repeated 10 times per second.

What the robot sees — and what it decides

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The saved card (`~/vla_one_step.png`): camera frame beside the numbers

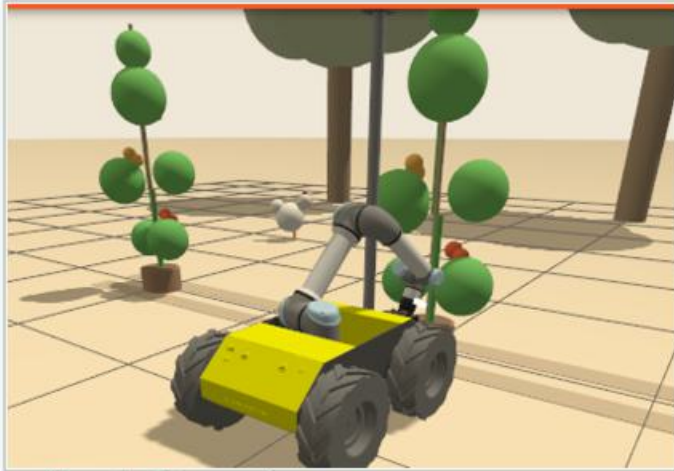


Left: the actual 224×224 wrist-camera frame the model received. Right: its inputs and output.

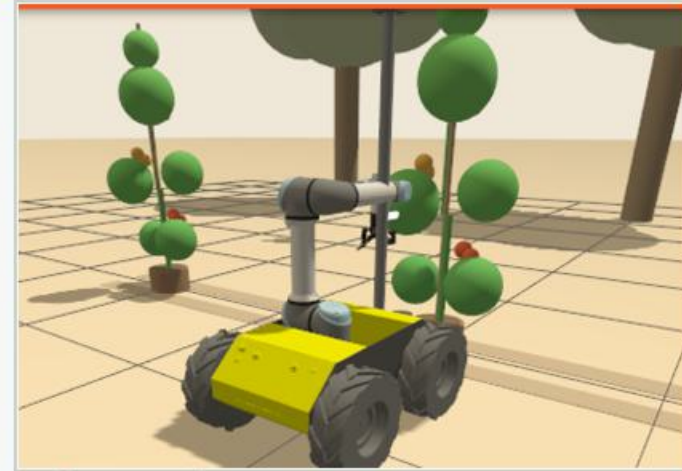
Step 5 · The pick — red tomato on the LEFT

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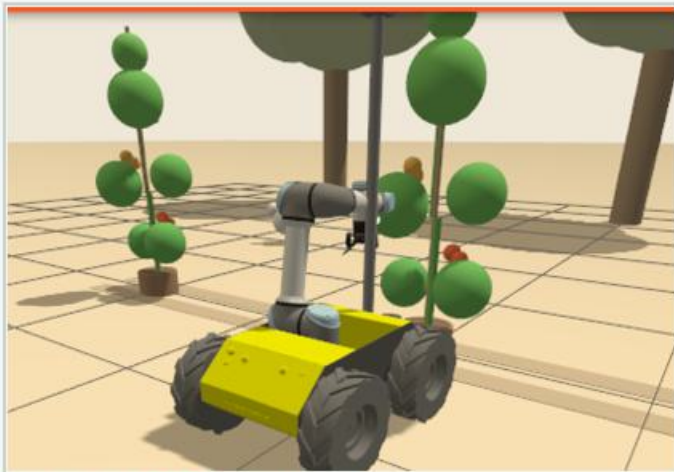
vla_d4 --task C --spawn --instruction "pick the red tomato"



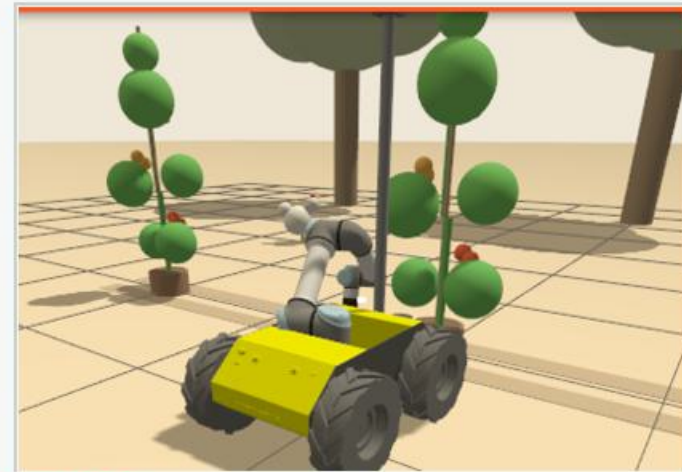
1 · scan above the left spot



2 · red seen → descend



3 · gripper closes on the red



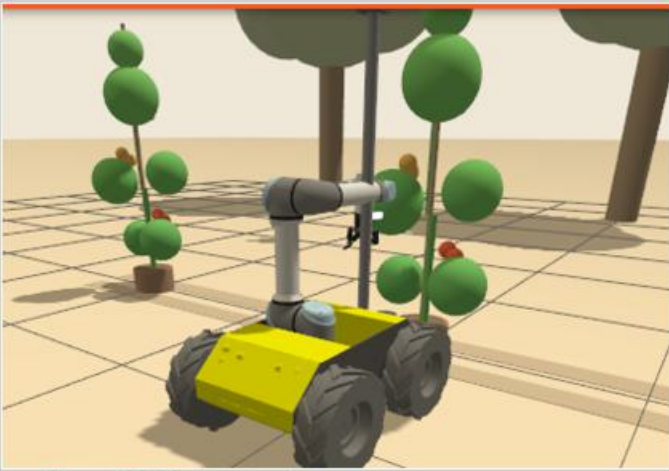
4 · hold — SUCCESS

SUCCESS in 35 model calls — judged by the grasp service naming the RED tomato, not by looks.

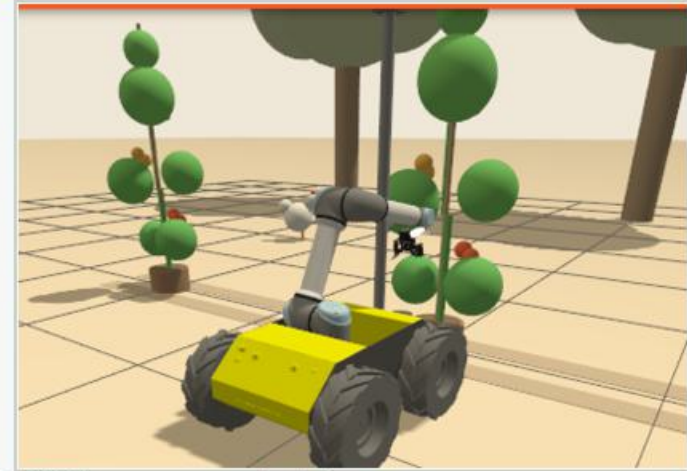
Active perception — red tomato on the RIGHT

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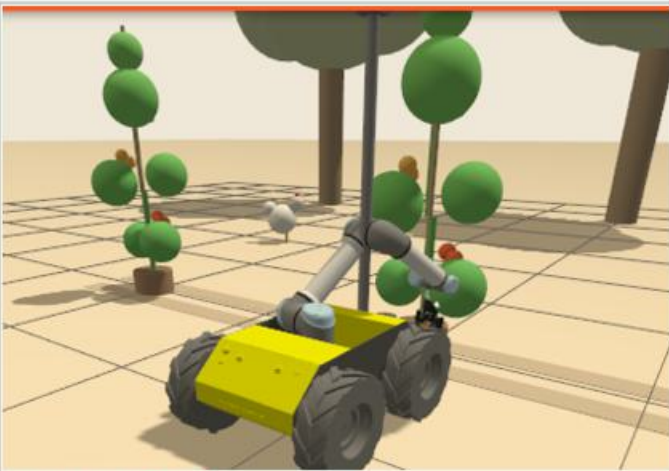
Same command, tomatoes swapped: watch it LOOK first



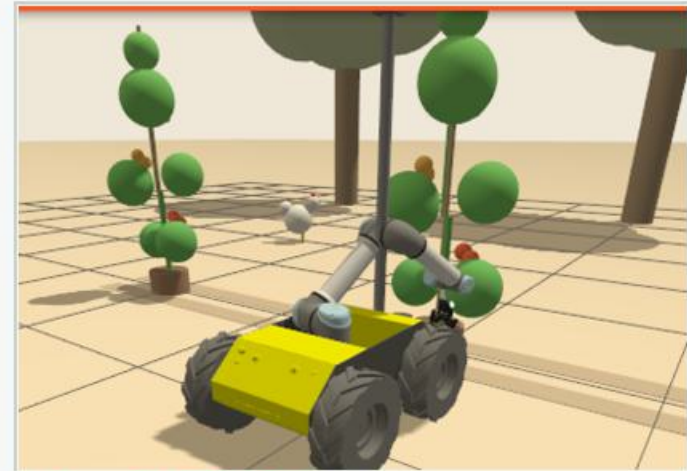
1 · scan above the LEFT spot...



2 · ...sees GREEN → pans right



3 · descends on the right



4 · picks the red — SUCCESS

Nobody programmed the scan — it was learned from demonstrations that contained it.

Measured, not vibes: 100/100

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The same rubric your own trained models are graded on

```
terminal 3 — the graded evaluation

$ ~/raise_venvs/lerobot/bin/python3 evaluator/evaluate.py --task C --trials 8

trial 1/8 red=L → success (mean act 7.2 ms)
trial 2/8 red=R → success (mean act 9.7 ms)
...
trial 8/8 red=R → success (mean act 8.2 ms)

— score —
success : 8/8 → 60/60
latency : max 167 ms (≤500) → 20/20
safety : 8/8 no-wrong-grab → 20/20
TOTAL : 100/100
```

8/8 correct picks alternating sides · 0 wrong grabs · max 167 ms per decision.

1. `git clone github.com/aniskoubaa/raise2026-student && ./start.sh`

installs everything, opens the sim

2. `get_brain_d4`

the trained model (or auto-downloads from Hugging Face on first run)

3. `vla_one_d4 --spawn`

one brain call, explained

4. `vla_d4 --task C --spawn --instruction "pick the red tomato"`

the pick you just watched

Open the step-by-step guide: `xdg-open lab_guides/lab2_handson.html`

New to ROS? the gentler one: `xdg-open lab_guides/lab2_beginner_handson.html`

Model card + weights: huggingface.co/scalexi/smolvla-raise2026-ripeness-ref